



Seamless Knitting

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Background

1.1 Context

Seamless knitting represents a highly integrated textile production technology where entire garments are produced directly on knitting machines, removing almost all cutting and sewing operations.

This innovation, (*in this case we base our model on the machines from Shima Seiki (WHOLEGARMENT®) and Santoni (SM8 series)*), achieves near-zero fabric waste and higher material efficiency compared to traditional knitwear production.

However, the process is more energy-intensive per kilogram of fabric due to complex knitting paths, longer cycle times, and higher frictional resistance, particularly with fine yarns.

Average machine characteristics (literature and datasheets):

- **Machine weight:** 750 - 1,400 kg
- **Installed power:** 1.5 - 2.2 kW (varies with gauge)
- **Knitting speed:** 1.3 - 1.6 m/s
- **Production rate:** 0.7 - 2.0 garments / hour
- **Energy range:** ~0.9 - 17 kWh kg⁻¹ garment

(Sources: Shima Seiki 2025; Knox 2011; Maity 2021; NTNU 2024)

1.2 Scope and Functional Unit

The seamless knitting model includes:

- **Direct electricity consumption** of the machine;
- **Indirect energy use** (compressors, HVAC, lighting, automation);
- **Lubricant use** and air emissions;
- **Capital goods and end-of-life allocation**;
- **Building infrastructure**.

The **functional unit** is **1 kg of knitted garment** produced under average industrial conditions.

Excluded: fibre and yarn production (modelled upstream), textile/material waste (yarn losses and waste-yarn treatment), which is handled externally by the caller via the process-specific IOR, building construction costs beyond the amortised area share modelled in Section 3.4, finishing, packaging and transport.

Background data: ecoinvent 3.12, system model: cut-off by classification. Background processes are linked via generic market activities (GLO/Row)

Note on sources: machine weight and energy range data are drawn from manufacturer datasheets (Shima Seiki, Santoni) and academic sources. Some supporting references (K&G Garment, Rakib Ahmed thesis) are commercial or grey literature and are used only for indicative benchmarking of material utilisation and yarn waste rates.

Methodological Overview

2.1 Yarn losses and lubricant - dependency on yarn fineness (dtex)

Concept

Yarn losses and lubricant demand depend on yarn fineness and machine efficiency.

Finer yarns require longer running time, higher tension control, and more lubricant per kilogram of output.

Base definition

"Input quantity refers to the net yarn required to produce 1 kg of knitted fabric. Yarn losses during the knitting process (thread breaks, edge waste, trial runs) should be included."

(Laursen et al., 2007; van der Velden et al., 2014; Sandin et al., 2019 – Mistra)

Empirical references

- **Material realisation:** 94 - 98 % (Maity et al., 2021; Rakib Ahmed 2024)
- **Lubricant:** 8 g kg⁻¹ fabric "Sandin et al.2019 pp45 - The lubricant was assumed to correspond to 8% of the weight of the knit."

Values applied

Parameter	Average	Comment
Yarn loss	2.5 %	Derived from 94–98 % yarn utilisation
Lubricant	0.008 kg kg ⁻¹	Upper-bound value adopted for high-load operation

2.2 Electricity consumption – direct and indirect energy

Direct electricity (machine operation)

Based on EcoBalyse, Maity et al. (2021) and Shima Seiki datasheets, seamless knitting machines typically consume **1.2–2.1 kWh kg⁻¹** of fabric.

A mid-range **average value of 1.5 kWh kg⁻¹** is adopted for modelling.

Indirect energy (compressors, HVAC, lighting, cleaning)

Shared methodology: [Indirect Energy in LCA — Shared Methodology \(Knitting\)](#).

The same structure used in flat and circular knitting is retained: **4.46 kWh/kg** (ADEME & Ministère de la Transition Écologique, 2024; $2.4 \div 0.35 \times 0.65 = 4.46$ kWh/kg), distributed as 35% machines, 25% compressors, 30% HVAC, 6.5% lighting, 3.5% auxiliaries. **4.46 kWh kg⁻¹**. Is a derivation the EcoBalyse upper estimate for knitting machine electricity is 2.4 kWh/kg (ADEME & Ministère de la Transition Écologique, 2024), representing 35% of total plant electricity based on the subsystem distribution above, and distributed as:

- 35 % knitting machines
- 25 % compressors
- 30 % HVAC
- 6.5 % lighting
- 3.5 % auxiliaries

This ensures methodological alignment across knitting technologies.

3. Capital Goods and End-of-Life

3.1 Modelling approach

Shared methodology: [Capital Goods & End-of-Life \(Machinery\) in LCA](#) | [Building Infrastructure in LCA](#). This section states only seamless-knitting-specific inputs.

Capital goods and end-of-life are included as mass-based foreground flows, consistent with EF guidelines and ecoinvent practice.

The machine mass is amortised over its lifetime production output:

$$m_{\text{machine per kg garment}} = \frac{M_{\text{machine}}}{Q_{\text{lifetime}}}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

3.2 Machine data

Parameter	Value	Source
Machine model	Santoni SM8-TR1S	Santoni (2023)
Machine mass	750 kg	Santoni (2023), SM8-TR1S datasheet
Productivity	~26.2 kg/h	Textileworld.com (2007); Maity et al. (2021)

3.3 Lifetime parameters — average scenario

Parameter	Value
Service life	15 years
Operating hours	16 h/day
Operating days	300 days/year
Productivity	26.2 kg/h
Q_lifetime	15 × 16 × 300 × 26.2 = 1,886,400 kg
Machine allocation	750 ÷ 1,886,400 = 0.000397 ≈ 0.0004 kg/kg

The ecoinvent exchange is expressed in units (1 unit = 1 machine of 2000 kg):

$$\text{Amount} = \frac{0.0004}{750} = 5.33 \times 10^{-7} \text{ unit/kg garment}$$

ecoinvent activity: `market for building machine {GL0} | ecoinvent 3.12 cut-off`

Scenario	Years	Hours/day	Days/year	Productivity (kg h ⁻¹)	Allocation (kg/kg garment)
Average	15	16	300	26.2	0.0004

3.4 End-of-life allocation

Machine material composition (Santoni, 2023; World Steel Association, 2021):

Material	Share	Mass (kg)	EoL per kg garment	ecoinvent activity	
Steel	85%	637.5 kg	637.5 ÷ 1,886,400 =	`market for waste steel {RoW}	cut-off`

Material	Share	Mass (kg)	EoL per kg garment	ecoinvent activity	
			0.000338 kg/kg		
Electronics / other	15%	112.5 kg	112.5 ÷ 1,886,400 = 0.0000596 kg/kg	`treatment of waste electric and electronic equipment, shredding {GLO}	cut-off`

Note on rounding: The summary table in Section 5 shows 0.0003 kg/kg and 0.0001 kg/kg respectively — these are conservatively rounded values. The precise figures above should be used for recalculation.

Material	Share (%)	Allocation (kg/kg garment)	Notes
Steel	85 %	0.0003	Recycled or disposed depending on scenario
Other materials	15 %	0.0001	Includes electronics and polymers

Machine composition:

Approx. 85% steel (637.5 kg) and 15% other materials (112.5 kg). Total: 637.5 + 112.5 = 750 kg

3.4 Building Infrastructure

Shared methodology: Building Infrastructure in LCA. This section states only seamless-knitting-specific inputs.

Process-specific value: **1.61E-05 m²/kg** — derived from Santoni SM8-TR1S envelope (2700 mm × 3150 mm ≈ 8.5 m²) × cell factor 3–4; consistent with openLCA Nexus organic cotton sweater benchmark. Seamless throughput per m² is intermediate between circular fabric knitting and flat/3D.

ecoinvent activity: **market for building, hall {GLO} | cut-off** (unit: m²) The building is represented by an ecoinvent-style construction dataset ("building, hall") whose reference flow is m² of industrial hall.

Allocation principle (area amortisation over lifetime output).

Building infrastructure is attributed to 1 kg of knitted garment by amortising the allocated hall area over the lifetime production associated with the knitting cell:

$$\text{Building intensity (m}^2\text{/kg)} = \frac{A_{\text{alloc}} \text{ (m}^2\text{)}}{Q_{\text{lifetime}} \text{ (kg)}}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

where A_{alloc} is the hall area allocated to the knitting cell (machine footprint + access + yarn storage + maintenance space + internal logistics), and Q_{lifetime} is the total knitted output produced during the reference service life under the chosen operating regime.

Technology rationale.

Seamless knitting (body-size knitting on large-diameter circular machines) produces complete garment tubes in a single operation, eliminating cut-and-sew waste. Its throughput per unit of floor space is intermediate between circular fabric knitting and flat or 3D knitting, as the larger machine envelope and more complex programming cycles reduce the production rate per m^2 relative to conventional circular knitting.

Practical implementation.

In the seamless knitting foreground unit process, we add an input: *building, hall* (unit = m^2) with amount = $1.61\text{E-}05 \text{ m}^2/\text{kg}$.

End-of-life of the building.

The building dataset includes construction, maintenance and decommissioning within the background system model. No separate building EoL module is created in the foreground; only the amortised m^2/kg input is provided.

Default value (this study). In the absence of site-specific A_{alloc} and Q_{lifetime} , the following reference intensity is applied, shared with the hosiery technology group given comparable machine scale and operating throughput:

Seamless knitting (garment): **$1.61\text{E-}05 \text{ m}^2/\text{kg}$**

Physical plausibility was cross-checked using machine envelope data from Santoni SM8-TR1S ($A = 2700 \text{ mm}$, $D = 3150 \text{ mm}$, envelope $\approx 8.5 \text{ m}^2$) with a cell factor of 3–4 $\times$. This value also aligns directly with the benchmark reported in the openLCA Nexus organic cotton sweater case study.

4. Waste and Material Efficiency

Traditional cut-and-sew processes typically lose **10 - 25 %** of fabric as trims (*GIZ. (2023). Sustainable apparel report - Cutting waste reduction. Deutsche Gesellschaft für Internationale Zusammenarbeit*).

Seamless knitting eliminates most cutting operations and achieves **> 95 % material utilisation**, corresponding to **< 5 % total waste** (K&G Garment 2025).

In line with the knitting convention, the foreground dataset does **not** include an inline textile-waste exchange. Instead, the 2.5% loss assumption is handled externally by the caller through the process-specific IOR, which scales upstream yarn input and routes the lost yarn to end-of-life treatment.

5. Summary of Inventory Parameters (Average Case)

Flow	Amount	Unit	Comment	Datasets
Product	1	kg	Average seamless knitted garment	
Lubricant	0.008	kg	Upper-bound for high-load production	lubricating oil production {RoW} cut-off
Electricity mix	1.5	kWh	Direct machine operation	market group for electricity, medium voltage {GLO} cut-off
Indirect energy	4.46	kWh	Compressors, HVAC, lighting, auxiliaries	market group for electricity, medium voltage {GLO} cut-off
Capital goods	0.0004	kg	Machine allocation (average)	market for building machine {GLO} cut-off
Building infrastructure	1,61E-05	m ²		market for building, hall {GLO} cut-off
EoL steel	0.0003	kg	Machine end-of-life	market for waste steel {RoW} cut-off
EoL others	0.0001	kg	Machine end-of-life	treatment of waste electric and electronic equipment, shredding {GLO} cut-off

Input–Output Ratio (IOR)

Definition: $IOR = 1 / (1 - L)$, where L is the yarn loss rate (as a fraction). For every 1 kg of finished knitted garment, the upstream yarn input must be scaled by IOR.

How the loss rate is derived

Seamless knitting eliminates most cutting and sewing operations, achieving >95% material utilisation. The 2.5% default loss rate reflects yarn breaks, machine setup waste, and start-up trials (Maity et al., 2021; Rakib Ahmed, 2024). The narrow range means a single default value is applied without BCS/WCS scenarios.

IOR value

Process	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per kg output (kg)	
Knitting, Seamless \	Knitting Process	2.5%	1.0256	1.0256	0.0256

How to apply: for 1 kg of seamless knitted garment, multiply upstream yarn input by **1.0256**. The 0.0256 kg of lost yarn is routed by the caller to **market for waste yarn and waste textile {Row}**. No inline yarn-loss or textile-waste exchange is included in the dataset itself.

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 Note for Gustavo — knitting convention change